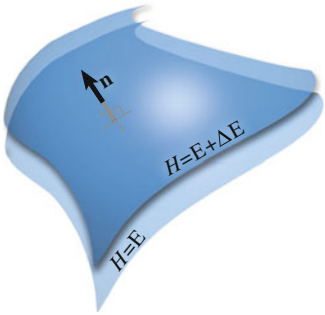


Fig. 5.1 A thickened shell between energy levels E and E + ΔE

Let us attempt to motivate the definition of the invariant measure. Consider a subset M of the level set Σ_{E}. We suppose $\|\nabla H\|$ to be bounded away from zero on Σ_{E}. At each point of M we let $\boldsymbol{n} = \nabla H/\|\nabla H\|$ be the unit normal to the energy surface and define a local coordinate ξ measuring distance along the direction of the unit normal, so that for a small offset $\Delta\mathrm{E}$ of the energy, $z + \xi\boldsymbol{n}$ is the point of intersection with the level set $\Sigma_{\mathrm{E}+\Delta\mathrm{E}}$. We thus obtain a thickened shell of points around the patch M with energy between E and E + ΔE (Fig. 5.1).

The volume of this thickened shell is clearly

$$\Delta\mathrm{V} = \int_{\mathcal{M}} \xi \mathrm{d}\sigma,$$

but since

$$\Delta\mathrm{E} = H(z + \boldsymbol{n}\xi) - H(z) = \nabla H \cdot \boldsymbol{n}\xi + \mathcal{O}(\Delta\mathrm{E}^2),$$

and $\boldsymbol{n} = \nabla H/\|\nabla H\|$, we have

$$\xi = \frac{\Delta\mathrm{E}}{\|\nabla H\|} + \mathcal{O}(\Delta\mathrm{E}^2),$$

and

$$\Delta\mathrm{V} = \int_{\mathcal{M}} \frac{\Delta\mathrm{E}}{\|\nabla H\|} \mathrm{d}\sigma + \mathcal{O}(\Delta\mathrm{E}^2).$$

Taking the ratio of ΔV to ΔE we find

$$\frac{\Delta\mathrm{V}}{\Delta\mathrm{E}} = \int_{\mathcal{M}} \frac{1}{\|\nabla H\|} \mathrm{d}\sigma + \mathcal{O}(\Delta\mathrm{E}).$$

The limit of this as $\Delta \mathrm{E} \to 0$ defines a quantity

$$\mu(\mathcal{M}) = \int_{\mathcal{M}} \frac{1}{\|\nabla H\|} \mathrm{d}\sigma.$$

As both volume and energy are invariant under the Hamiltonian flow, $\mu(\mathcal{M})$ must be invariant under the flow, thus

$$\mu(\mathscr{F}_t(\mathcal{M})) = \mu(\mathcal{M}).$$

The invariant measure of the whole level set Σ_{E} is defined as the *microcanonical partition function*

$$Z_{\mathrm{E}} = \mu(\Sigma_{\mathrm{E}}).$$

Example 5.1 Consider the harmonic oscillator $H(q,p) = p^2/2 + \Omega^2 q^2/2$. The energy surface $H = \mathrm{E} > 0$ is an ellipse. The partition function is obtained by integrating in the standard way. Parameterizing the surface by q, we have $p = \sqrt{2\mathrm{E} - \Omega^2 q^2}$, then

$$\begin{aligned}
\mu(\Sigma_{\mathrm{E}}) &= 4 \int_0^{\sqrt{2\mathrm{E}}} \frac{\sqrt{1 + p_q^2}}{\sqrt{H_p^2 + H_q^2}} \mathrm{d}q \\
&= 4 \int_0^{\sqrt{2\mathrm{E}}/\Omega} \frac{\sqrt{1 + \frac{\Omega^4 q^2}{2\mathrm{E} - \Omega^2 q^2}}}{\sqrt{p^2 + \Omega^4 q^2}} \mathrm{d}q \\
&= 4 \int_0^{\sqrt{2\mathrm{E}}/\Omega} \frac{1}{\sqrt{2\mathrm{E} - \Omega^2 q^2}} \mathrm{d}q \\
&= 2\pi/\Omega.
\end{aligned}$$

Example 5.2 Consider a system of N non-interacting particles of mass m moving in periodic boundary conditions within a box of volume V. The Hamiltonian is

$$H = \frac{1}{2m} \sum_{i=1}^{\mathrm{N}} \|\boldsymbol{p}_i\|^2$$

so

$$Z_{\mathrm{E}} = \mathrm{V}^{\mathrm{N}} \int_{\mathrm{S}} \frac{m}{r} \mathrm{d}\sigma$$